Aga Khan University Examination Board

Notes from E-Marking Centre on HSSC-I Physics Examination April/ May 2019

Introduction

This document has been produced for the teachers and candidates of Higher Secondary School Certificate (HSSC) Part I Physics. It contains comments on candidates' responses to the 2019 HSSC-I Examination, indicating the quality of the responses and highlighting their relative strengths and weaknesses.

E-Marking Notes

This includes overall comments on candidates' performance on every question and *some* specific examples of candidates' responses which support the mentioned comments. Please note that the descriptive comments represent an overall perception of the better and weaker responses as gathered from the e-marking session. However, the candidates' responses shared in this document represent some specific example(s) of the mentioned comments.

Teachers and candidates should be aware that examiners may ask questions that address the Student Learning Outcomes (SLOs) in a manner that requires candidates to respond by integrating knowledge, understanding and application skills they have developed during the course of study. Candidates are advised to read and comprehend each question carefully before writing the response to fulfil the demand of the question.

Candidates need to be aware that the marks allocated to the questions are related to the answer space provided on the examination paper as a guide to the length of the required response. A longer response will not in itself lead to higher marks. Candidates need to be familiar with the command words in the SLOs which contain terms commonly used in examination questions. However, candidates should also be aware that not all questions will start with or contain one of the command words. Words such as 'how', 'why' or 'what' may also be used.

General Observations

This year candidates performed well on questions related to dimensions, fluid dynamics, impulse, SHM, speed of sound, polarisation and uniform circular motion. However, candidates did not perform well in questions based on equilibrium, work done in gravitational field, word problems based on thermodynamics and had difficulty in articulating their answers to reasoning and discussion type questions.

Detailed Comments:

Constructed Response Questions (CRQs)

Question 1:

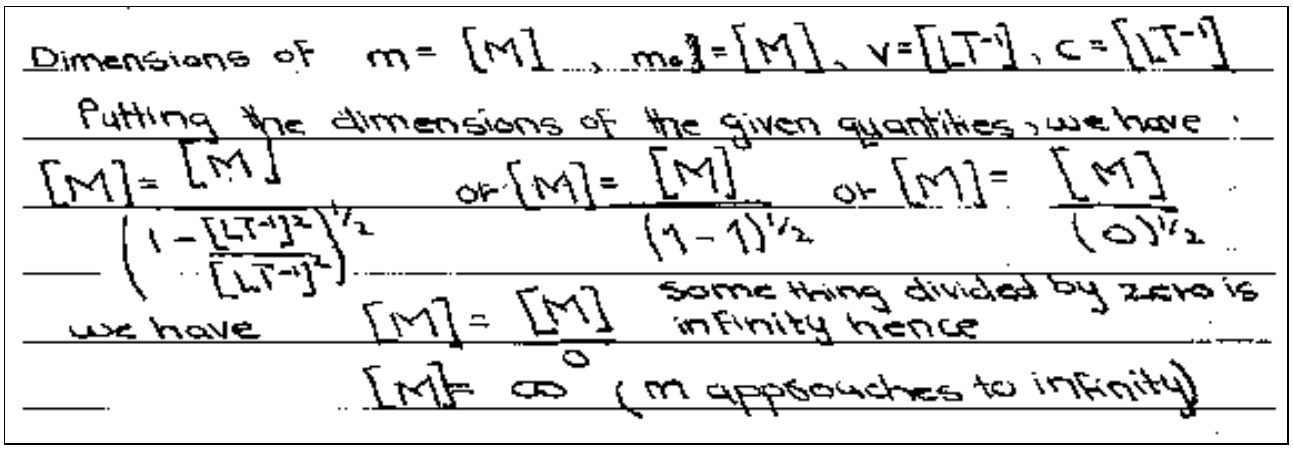
The consequence of special theory of relativity gives the following relation of an object moving with a speed 'v' approaching to the speed of light 'c', relativistic mass 'm' and rest mass 'm_o'.

$$m = \frac{m_o}{\left(1 - \frac{v^2}{c^2}\right)^{1/2}}$$

By using dimensions, show that 'm' in the given expression approaches infinity ($m \to \infty$).

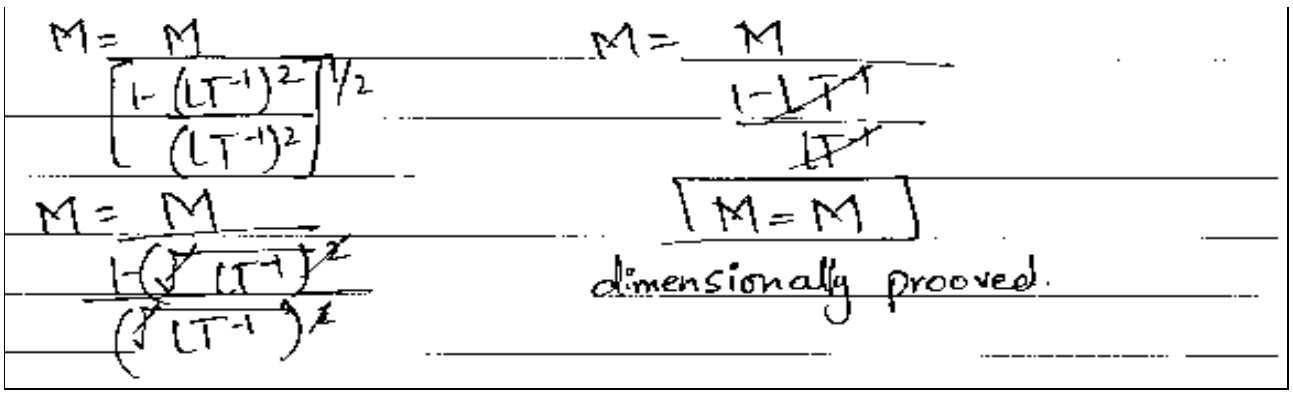
Better responses correctly showed that 'm' in the given expression approaches infinity ($m \to \infty$). High-scoring candidates wrote the correct dimension of mass and velocity and showed correct calculations to achieve zero in the denominator. Hence, they proved that mass approaches to infinity.

Example:



Weaker responses showed candidates' lack of understanding about the dimension. Few low-scoring candidates easily substituted the dimension of mass and velocity in the given relation but made mistakes in calculation.

Example:



Question 2:

Consider the given situations having no acceleration.

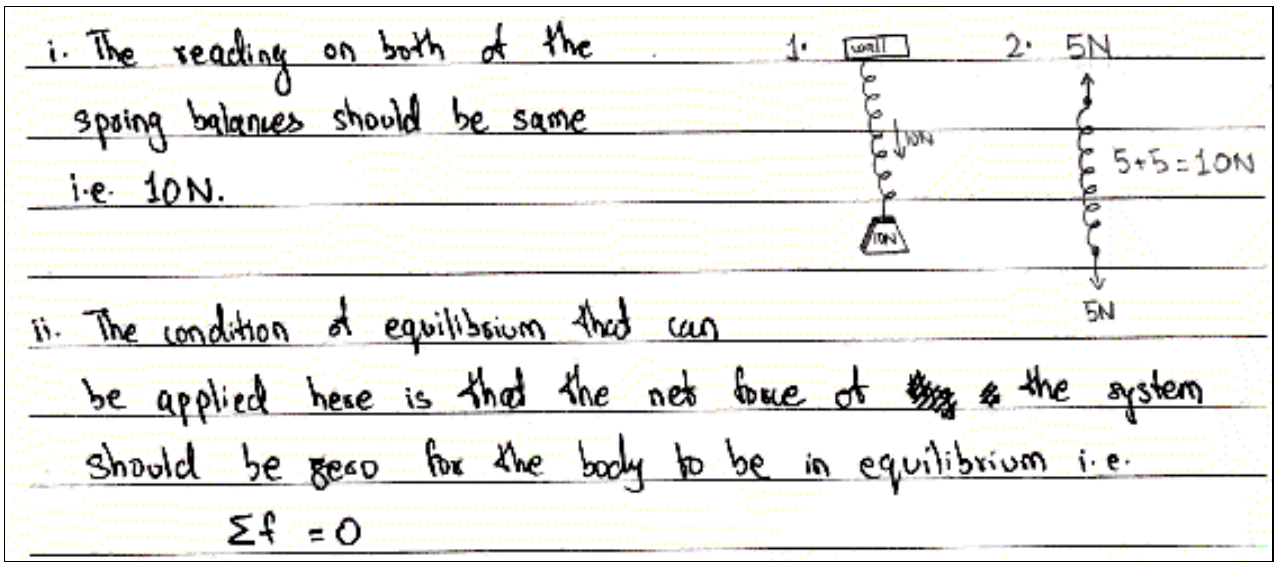
Situation 1: A man pulls a spring balance attached to a fixed wall with a force of 10 N.

Situation 2: Two different ropes are attached to each end of a spring balance. Two men pull the spring balance by applying a force of 5 N at each end in opposite directions.

i. Identify whether the reading on the spring balance in situation 1 is greater than, less than or equal to the reading on the spring balance in situation 2?
ii. Identify the condition of equilibrium based on your answer in part (i). Justify your answer.

In better responses, candidates correctly identified both cases by analysing the given situations. In first part, candidates wrote that the reading on the spring balance in situation 1 is equal to the reading in situation 2. In the second part, candidates correctly identified the first condition of equilibrium and justified that the vector sum of all the forces acting on an object in equilibrium is zero.

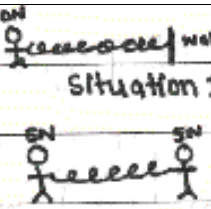
Example:



Weaker responses revealed candidates' lack of understanding and in-depth knowledge about the concept of equilibrium. Low-scoring candidates confused in identifying the first and second condition of equilibrium. They correctly analysed both the situations but were unable to justify their answers or failed to explain the condition of equilibrium.

Example:

i) The reading on the spring balance in situation 1 is equal to the reading on the spring balance in situation 2.



ii) The given condition of equilibrium is static because there is no acceleration in the spring balance. Both mans are at rest no velocity is applied or displacement is changed.

Question 3:

Prove that rate of change of linear momentum of a body is equal to the net force acting on the body.

Better responses proved that rate of change of linear momentum of a body are equal to the net force acting on the body. High-scoring candidates analysed the question properly and applied second law of motion. They correctly made the comparison, i.e. force is equal to the rate of change of momentum and mentioned the following mathematical steps in the derivation.

$a = \frac{v_f - v_i}{\Delta t}$, $a = \frac{F}{m}$, $\frac{F}{m} = \frac{v_f - v_i}{\Delta t}$, $F = \frac{mv_f - mv_i}{\Delta t}$, $F = \frac{P_f - P_i}{\Delta t}$, $F = \frac{\Delta P}{\Delta t}$

Example:

Let $F = ma \rightarrow (1)$.

$\therefore a = \frac{\Delta v}{\Delta t}$

Substituting value of a in eq (1)

$F = m\left(\frac{\Delta v}{\Delta t}\right) \rightarrow (ii)$

$\therefore \Delta v = v_f - v_i$

Substituting in eq (ii).

$F = \frac{m(v_f - v_i)}{\Delta t}$

$F = \frac{mv_f - mv_i}{\Delta t}$

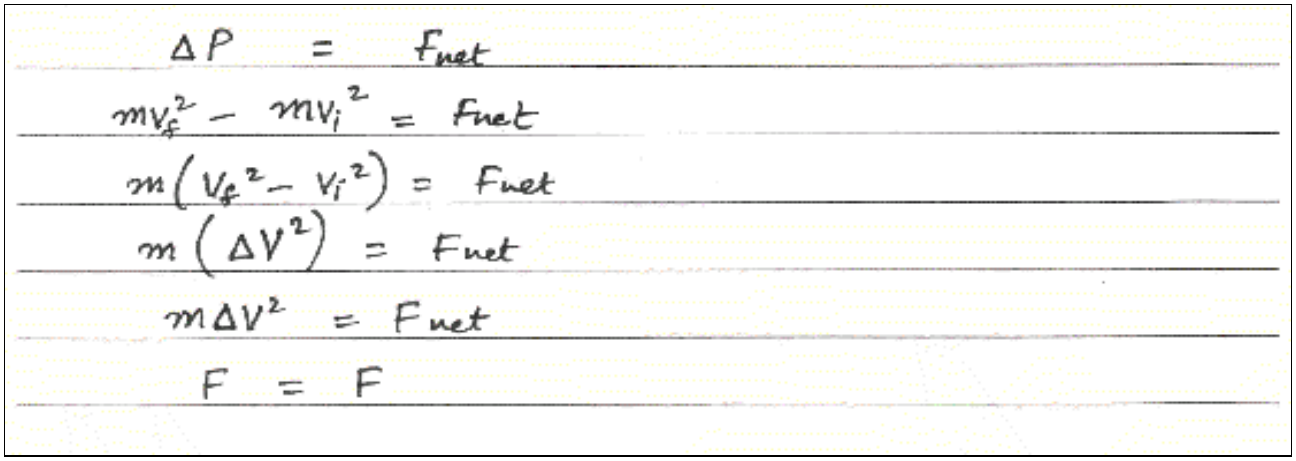
$\therefore P = mv$ = momentum

$F = \frac{P_f - P_i}{\Delta t}$.

*Hence it's proved that rate of change of linear momentum is equal to net force acting on body.

Weaker responses reflected that candidates were unable to prove the given condition. Some of the candidates made the relationship between momentum and applied force and finally, provided the wrong evidence of the relation, i.e. Applied force (F) = Applied force (F). Few other candidates paraphrased the question without mathematical proof.

Example:



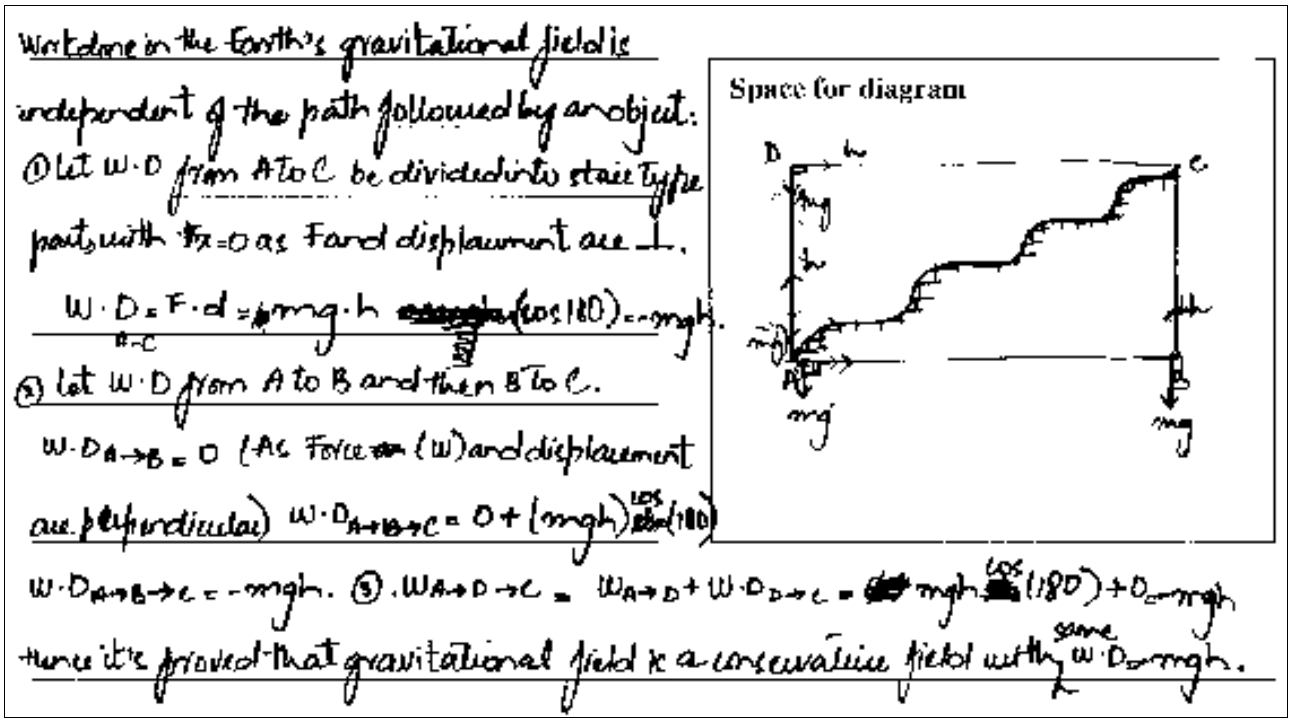
Question 4:

Work done in the Earth's gravitational field is independent of the path followed by an object.

Explain the given statement. Draw a diagram to support your explanation.

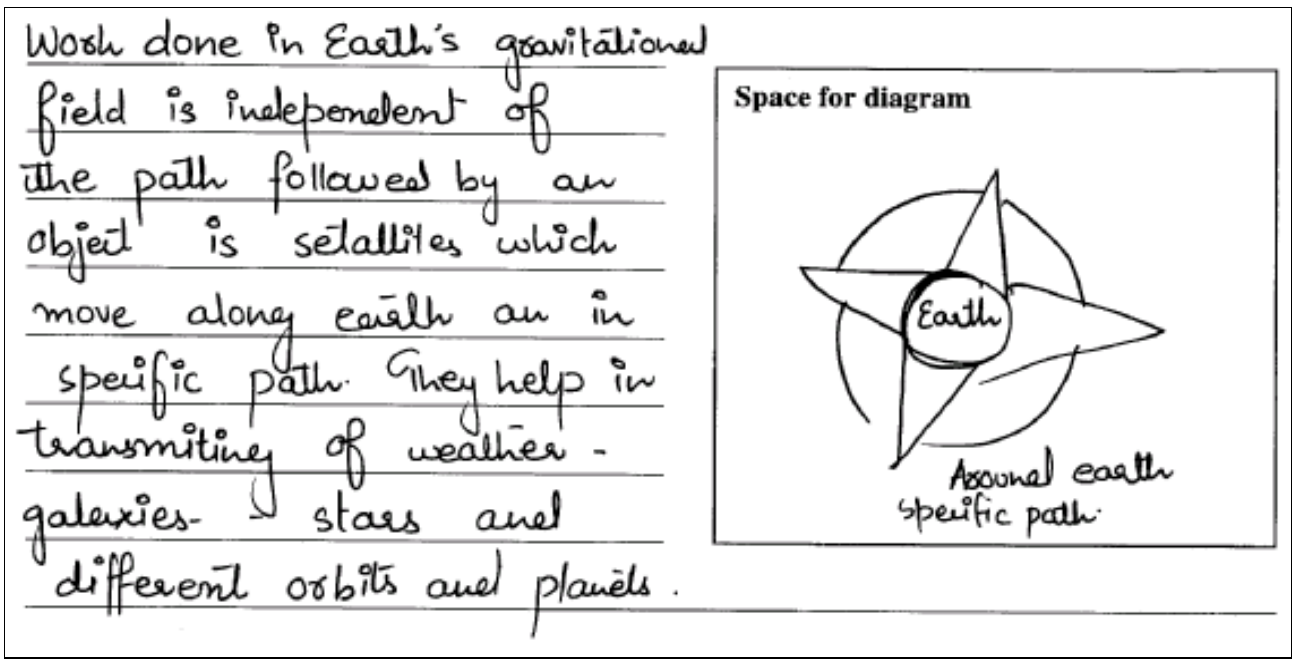
Most of the candidates were not able to perform well in this question. However, a few high-scoring candidates explained that the work done in the Earth's gravitational field is independent of the path followed by an object. Furthermore, they explained that gravitational field is the conservative field and drew three different paths in the labelled diagram alongwith the mathematical evidences to prove that work done is independent of the path followed by an object.

Example:



Weaker responses presented incomplete or partially correct explanation. In many cases, low-scoring candidates wrote that Earth's gravitational field is independent of the path followed by a satellite which was irrelevant to the demand of the question.

Example:



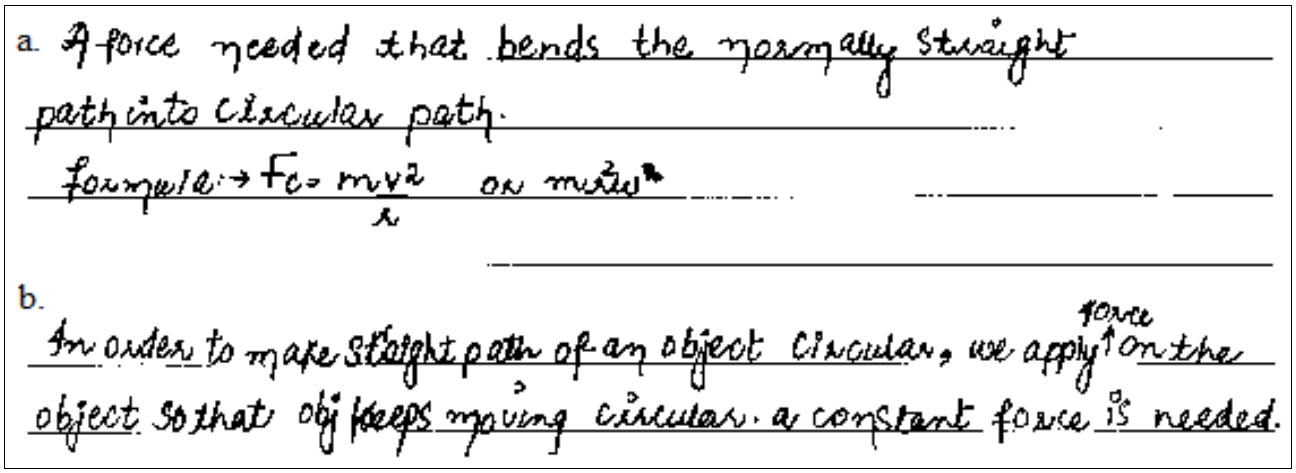
Question 5:

a. Define centripetal force and write its mathematical formula.

b. Why must centripetal force be applied on an object to move it in a circular path?

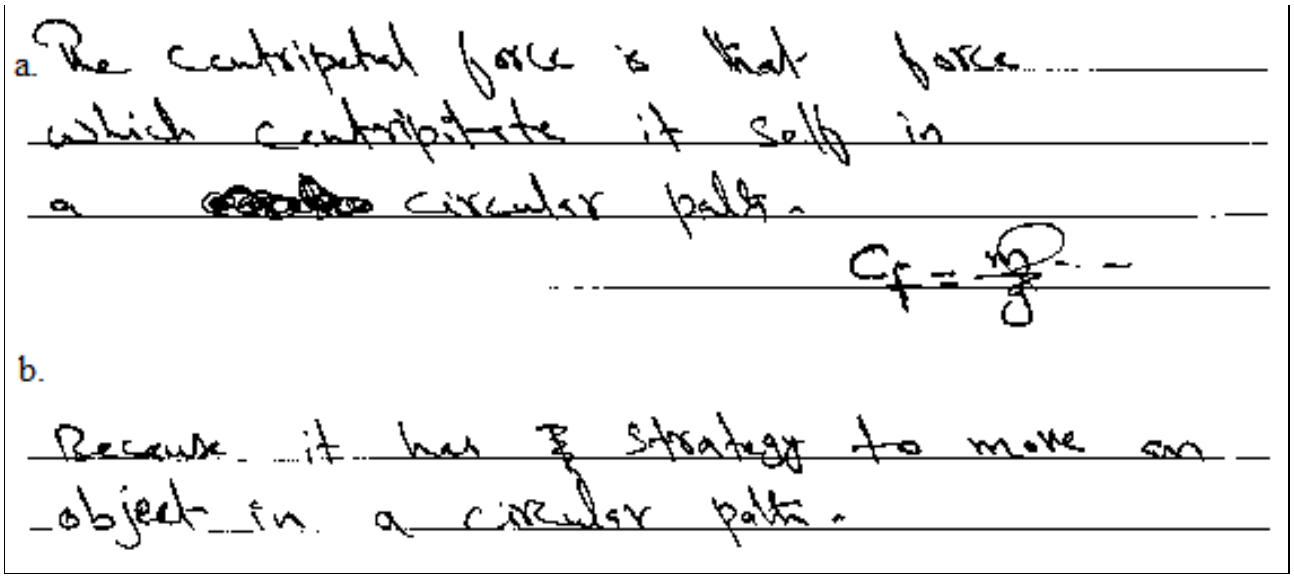
This question was generally well-attempted by majority of the candidates. In *better responses*, candidates wrote the correct definition of centripetal force with its correct mathematical formula in part (a). In part (b), most of the high-scoring candidates correctly justified that the centripetal force must be applied on an object to move it in a circular path and the uniform circular motion of an object is always under the continuous influence of a force that changes the direction of the velocity of the object and hence, produces the acceleration in the object.

Example:



Weaker responses neither wrote the correct definition of centripetal force nor gave its mathematical formula. In these cases, it is advisable that subject teachers must focus on definitions of some important terminologies to clarify the basic points.

Example:

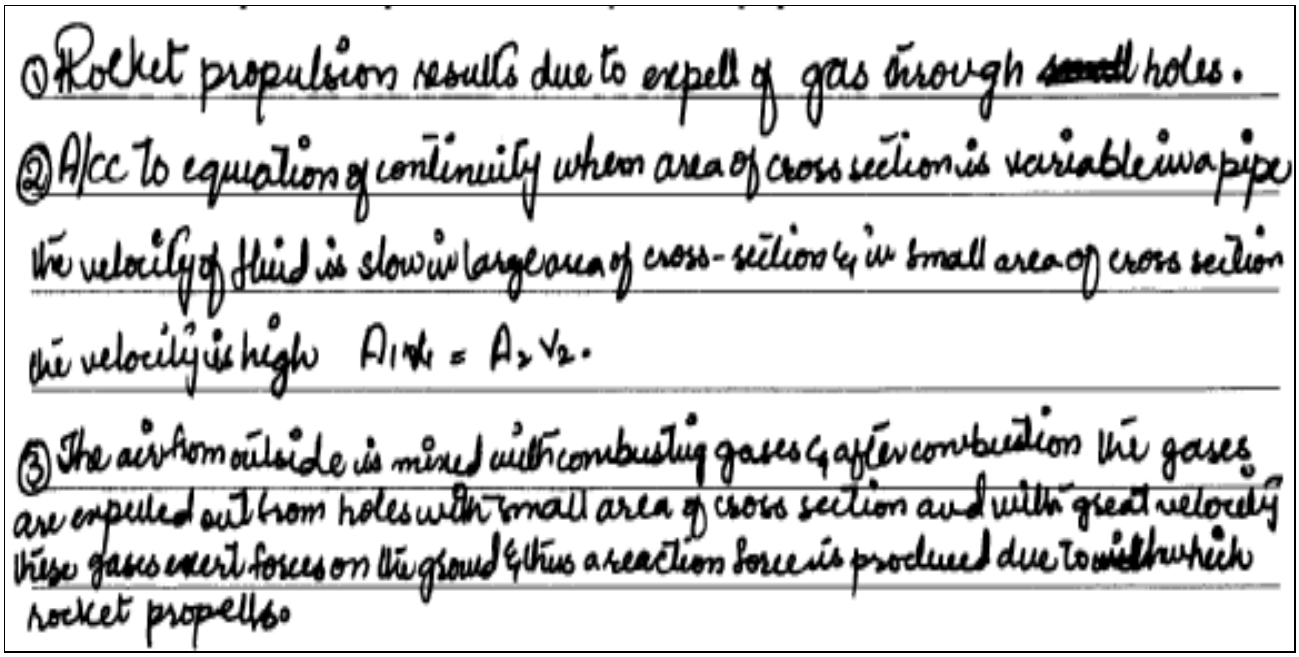


Question 6:

Relate in THREE points the equation of continuity with the propulsion of a rocket.

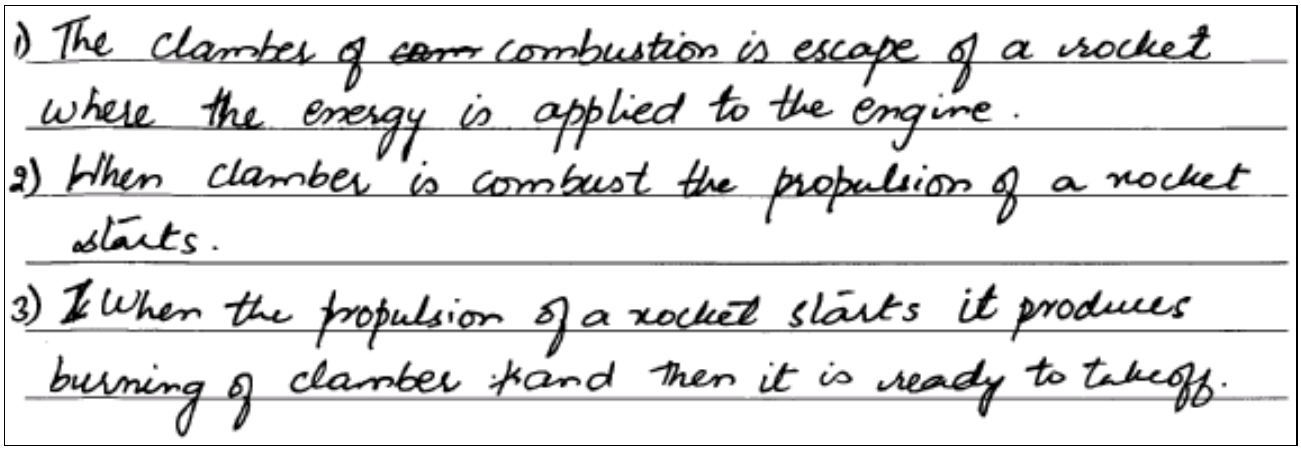
Better responses correctly related in three points the equation of continuity with the propulsion of a rocket. High-scoring candidates wrote that in a rocket, the area of cross-section for the ejection of exhaust gases is made small and according to the equation of continuity, the speed of gases increases and this leads to a raise in the momentum of the rocket which causes the force on the rocket to increase and thus, it flies fast. Some of the candidates also wrote the mathematical relation of the equation of continuity as $A_1V_1 = A_2V_2$.

Example:



Weaker responses demonstrated that candidates were unable to relate the equation of continuity with the propulsion of a rocket. Some low-scoring candidates wrote about the combustion chambers used in the rocket that provided the necessary force to the engine of a rocket to escape the Earth's gravitational field. Such statements were irrelevant to the demand of the question.

Example:



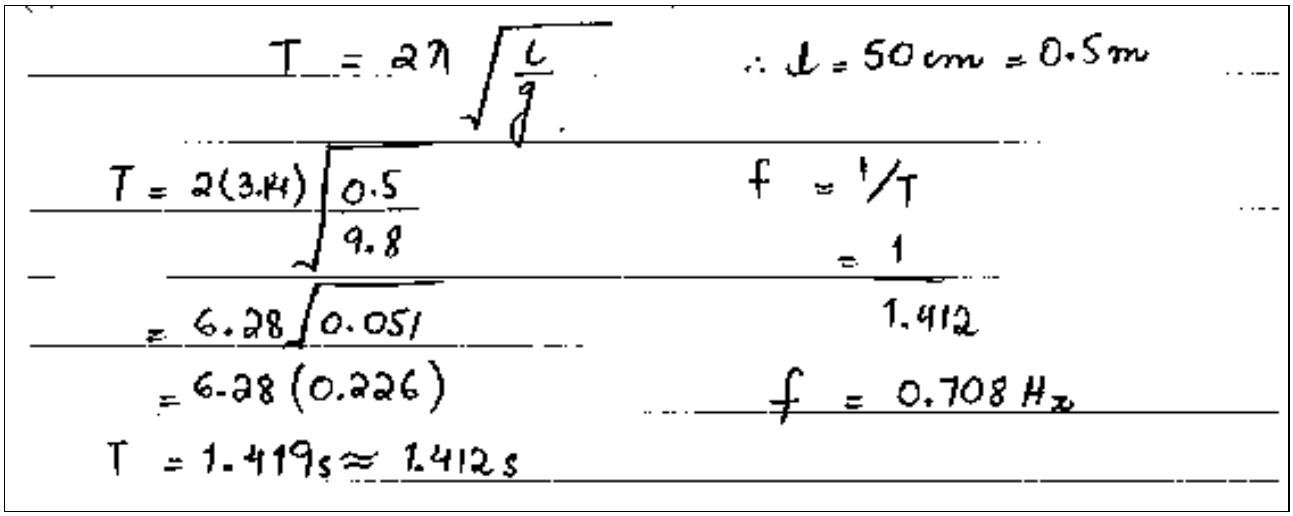
Question 7:

Calculate the frequency of a simple pendulum, if its length is 50 cm.

(**Note**: The acceleration due to gravity 'g' is 9.8 m/s^2.)

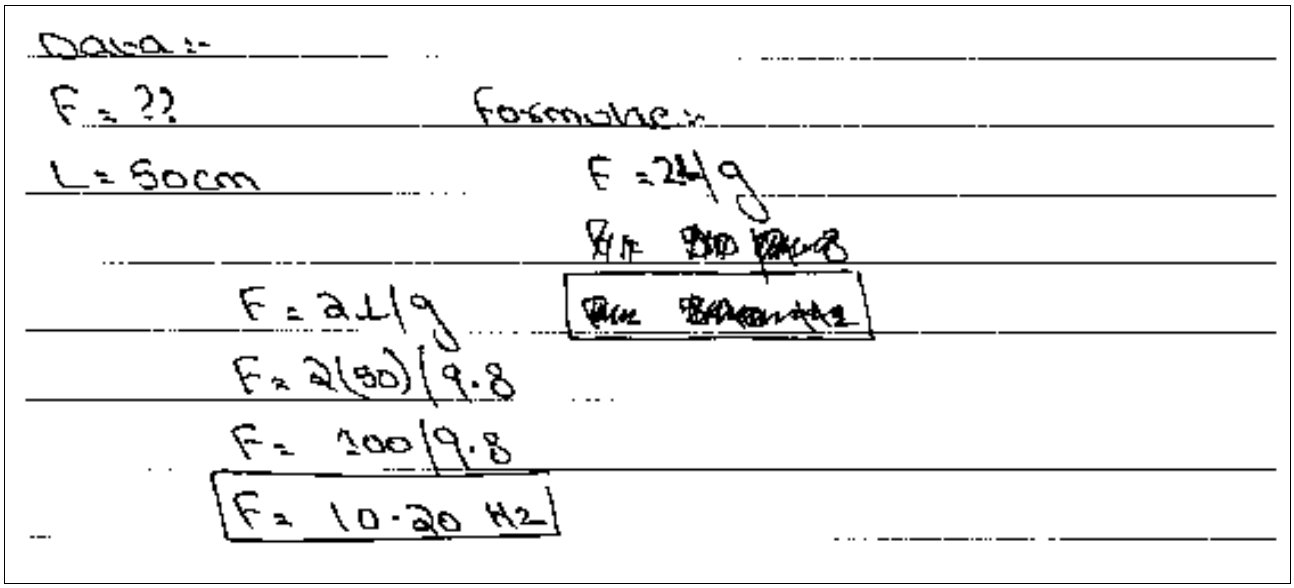
This question was generally well-attempted by the candidates. *Better responses* correctly calculated the frequency of a simple pendulum by converting 'cm' into 'm', replacing frequency 'f' with the reciprocal of the time period '1/T' and substituting the values in the formula $T = 2\pi\sqrt{\frac{l}{g}}$.

Example:



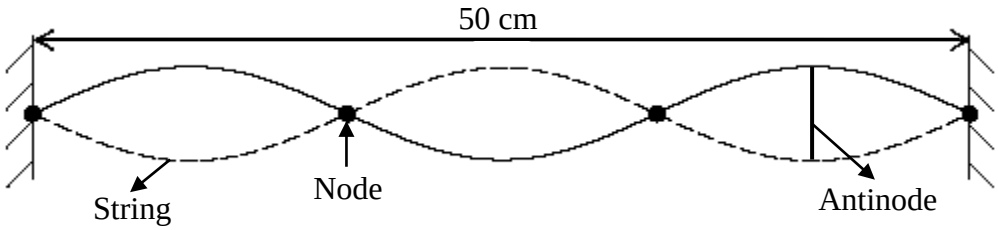
Weaker responses depicted that candidates were unable to write the correct formula of the time period of a simple pendulum. Few low-scoring candidates wrote about the relationship of the area of pipe which was irrelevant to the demand of the question. They even were not able to extract the correct data from the given question.

Example:



Question 8:

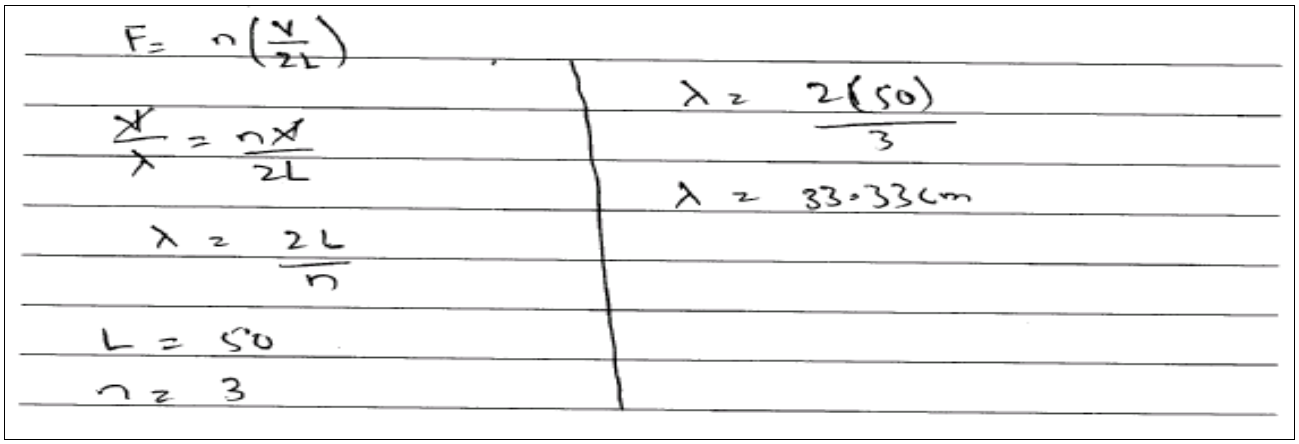
Standing waves are formed in an inextensible vibrating string as shown in the given diagram.



Find the wavelength of the standing wave.

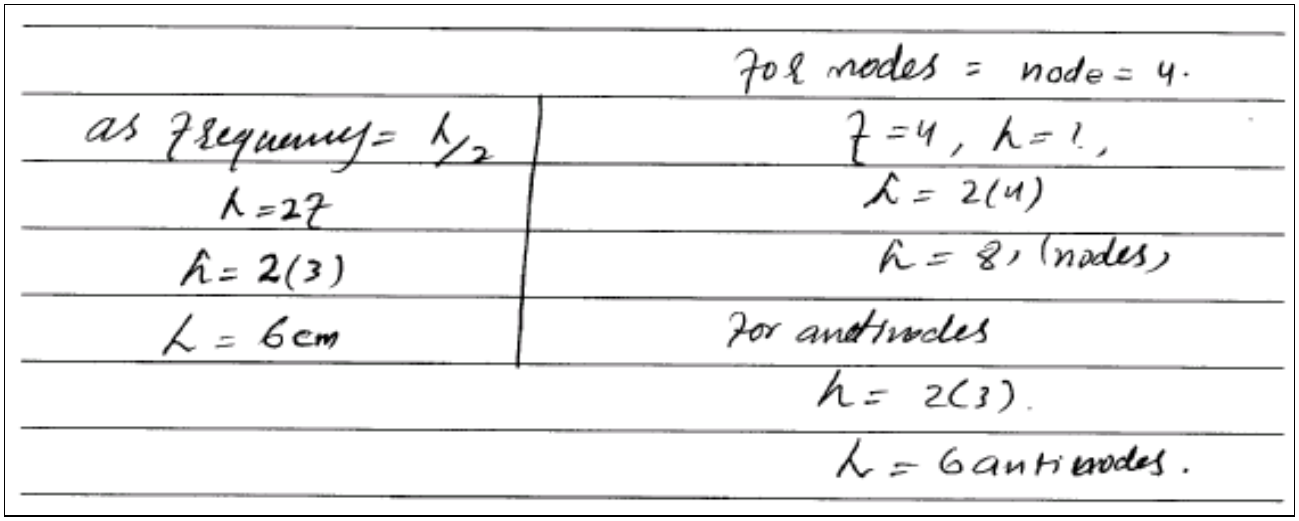
This question was generally well-attempted by the candidates. *Better responses* showed good command over the standing wave and their interpretation. High-scoring candidates extracted the correct data from the ray diagram of the standing wave, converted 'cm' into 'm' and used the correct formula for finding the wavelength, i.e. $L = n\frac{\lambda}{2}$.

Example:



Weaker responses revealed that candidates were unable to understand the demand of the question, and calculated the time period by taking the frequency of 50 Hz. They failed to identify the length of the standing wave as well. Most of the candidates wrote wrong formula for finding the wavelength.

Example:



Question 9:

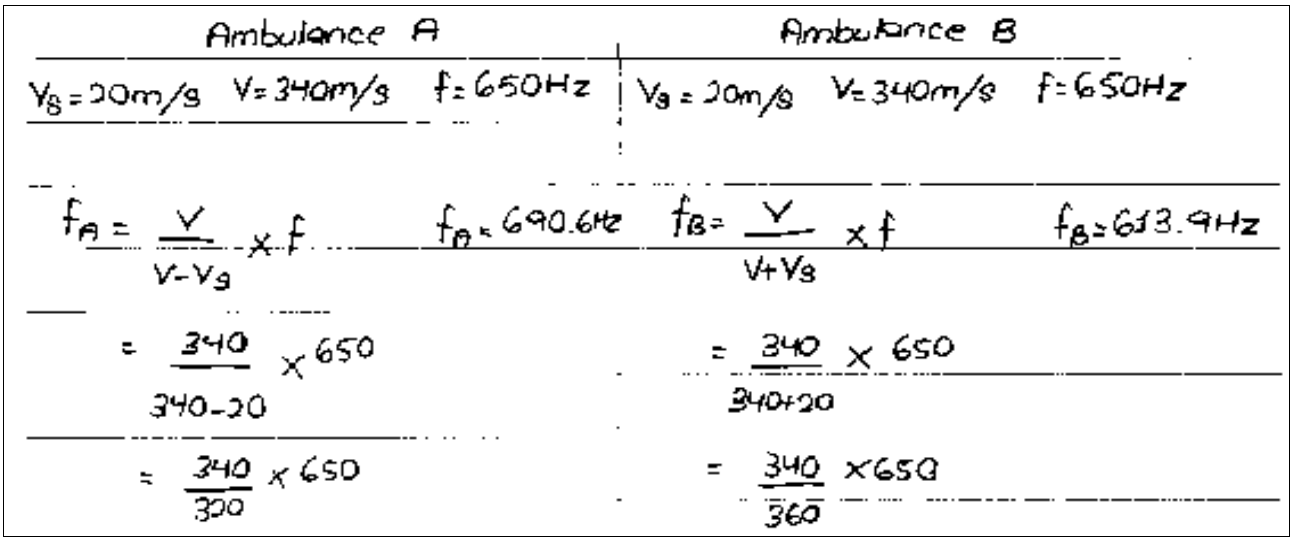
Two ambulances ‘A’ and ‘B’ are moving with velocity of 20 m/s with normal siren frequency of 650 Hz. Ambulance ‘A’ moves towards a stationary listener while ambulance ‘B’ moves away from the stationary listener.

Calculate the frequency of siren heard by the stationary listener from each ambulance ‘A’ and ‘B’.

(**Note**: The speed of sound in the air is 340 m/s.)

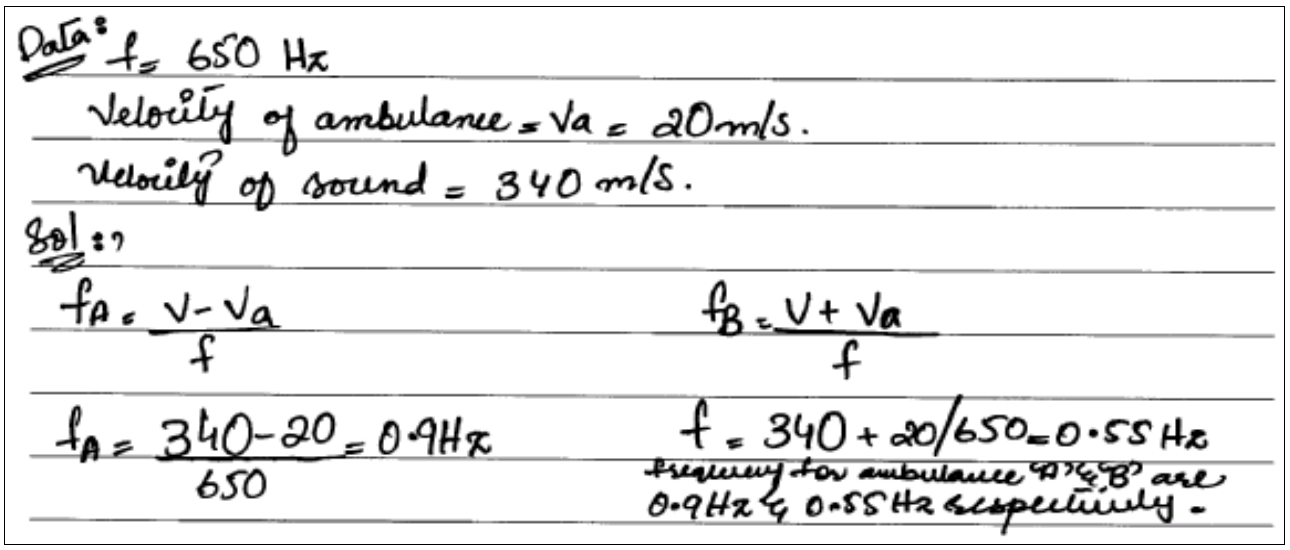
Better responses calculated the frequency of siren heard by the stationary listener from each ambulance 'A' and 'B'. The candidates correctly applied the two different formulae of the apparent frequency of the sound listened by the stationary listener due to ambulances 'A' and 'B', i.e. $f_A = \left(\frac{v}{v-u_s}\right)f$ and $f_B = \left(\frac{v}{v+u_s}\right)f$ that showed their command over the understanding of all the cases of Doppler's effect.

Example:



In weaker responses, candidates were not able to properly articulate their answers. Most of the low-scoring candidates wrote the wrong formulae of the apparent frequencies of the sound listened by the stationary listeners due to both ambulances and ultimately, substituted the wrong values and got incorrect answers.

Example:

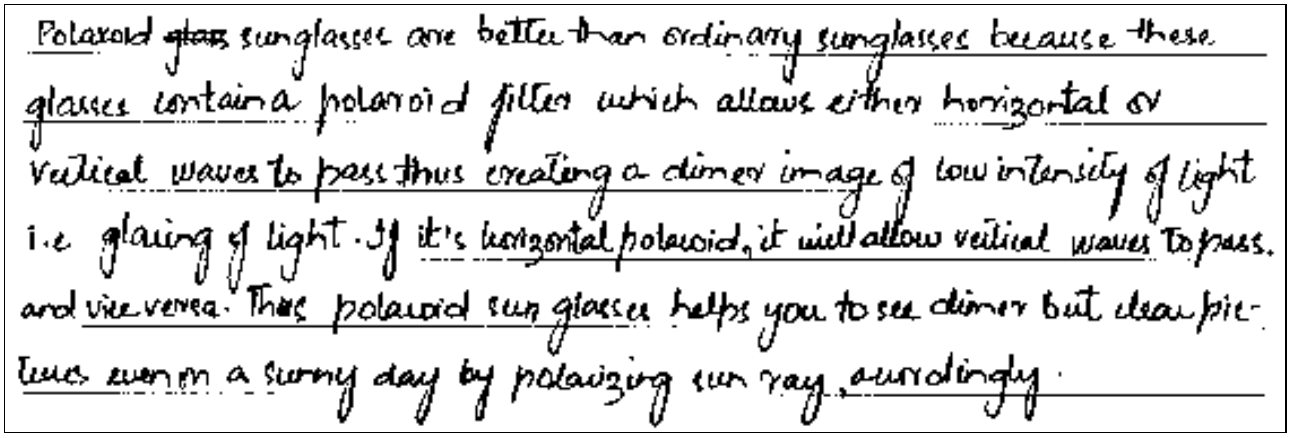


Question 10:

The polaroid sunglasses are better than ordinary sunglasses. Justify the given statement with reference to the glaring of light.

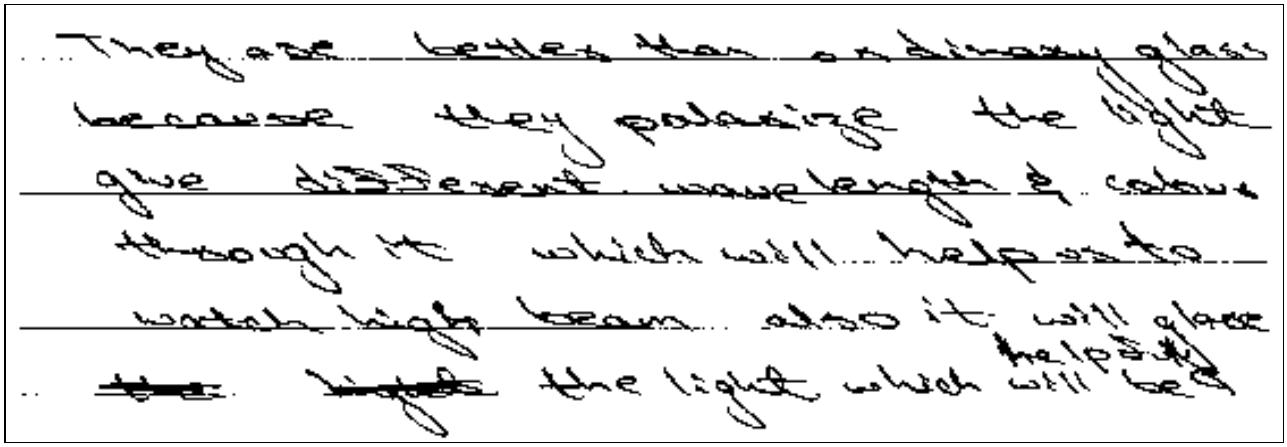
This question was generally well-attempted by the candidates. *Better responses* correctly justified that the polaroid sunglasses are better than ordinary sunglasses with reference to the glaring of light. High-scoring candidates explained the phenomenon of the polarisation as the sunlight which when reflected from water, glass and rough road surfaces with the large angles of incidences is partially polarised and produces a glare. Some of the candidates wrote that glare can be reduced by using polaroid sunglasses because they decrease the intensity of light passing through them. Few candidates explained that the polaroid sunglasses are better than ordinary sunglasses because the ordinary sunglasses cannot provide much less intensity of light.

Example:


Polaroid ~~glass~~ sunglasses are better than ordinary sunglasses because these glasses contain a polaroid filter which allows either horizontal or vertical waves to pass thus creating a dimer image of low intensity of light i.e glaring of light. If it's horizontal polaroid, it will allow vertical waves to pass. and vice versa. Thus polaroid sun glasses helps you to see dimer but clear pictures even on a sunny day by polarizing sun ray accordingly.

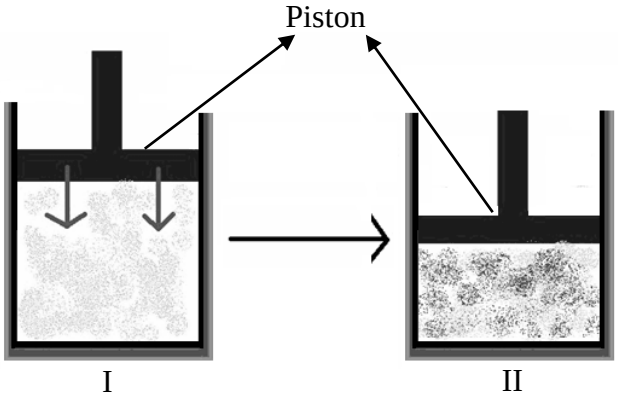
Weaker responses exhibited that candidates were unable to explain the worth of polaroid sunglasses over ordinary sunglasses. Such responses showed that candidates need more focus and attention on this topic. In some responses, candidates wrote the phenomena of Doppler's effect and its different cases which were not the demand of the question.

Example:


They are better than ordinary glass because they polarize the light give different wavelength & colours through it which will help us to watch high beam also it will glare the light which will be helpful

Question 11:

The given diagrams show a process in which the piston is compressed so that there is no heat exchange.



a. Identify the thermodynamic process illustrated in the given diagrams.
b. Discuss the relationship of the first law of thermodynamics for the process shown in the given diagrams when there is no heat transfer using mathematical equations.
c. In the illustrated process, will the temperature and internal energy raise or lower down?

This question was generally well-attempted by the candidates. In *better responses*, candidates correctly identified the thermodynamic process to be adiabatic process in part (a). In part (b), candidates discussed the relationship of the first law of thermodynamics for the process shown in the given diagrams with mathematical equations as adiabatic process is a process in which there is no heat transfer into or out of the system and $\Delta Q = 0$, $\Delta Q = \Delta U + \Delta W$, $0 = \Delta U + \Delta W$, $\Delta U = -\Delta W$. In part (c), high-scoring candidates illustrated the adiabatic process as a process during which temperature and internal energy are raised.

Example:

a. Adiabatic process. where there is no heat exchange.

b.

$\Delta Q = \Delta U + W$

So as no heat exchange $\Delta Q = 0$

$\Delta Q = 0$ so $0 = \Delta U + W$ $-\Delta U = W$

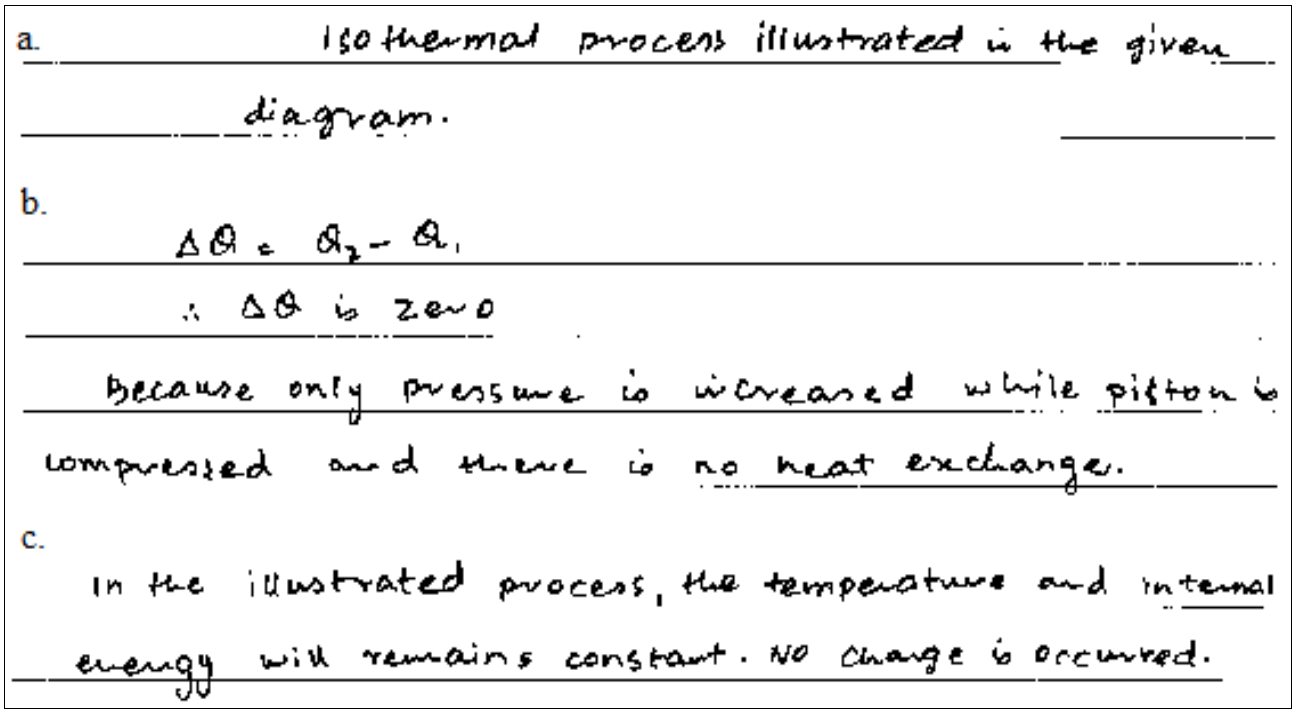
$-\Delta U = W$

c.

In compressed adiabatic process temperature and internal energy raises. (Increase in internal energy & temperature)

Weaker responses showed candidates' inability to identify the thermodynamic process. They identified it as isothermal or isobaric process in part (a). In part (b), low-scoring candidates just showed change in internal energy as $\Delta Q = Q_2 - Q_1$. In part (c), candidates incorrectly wrote that during the thermodynamics process, temperature and internal energy will remain constant.

Example:



Extended Response Questions (ERQs)
These questions offered a choice between part **a** and **b**

Majority of the candidates attempted question **12a** and they scored well in this part.

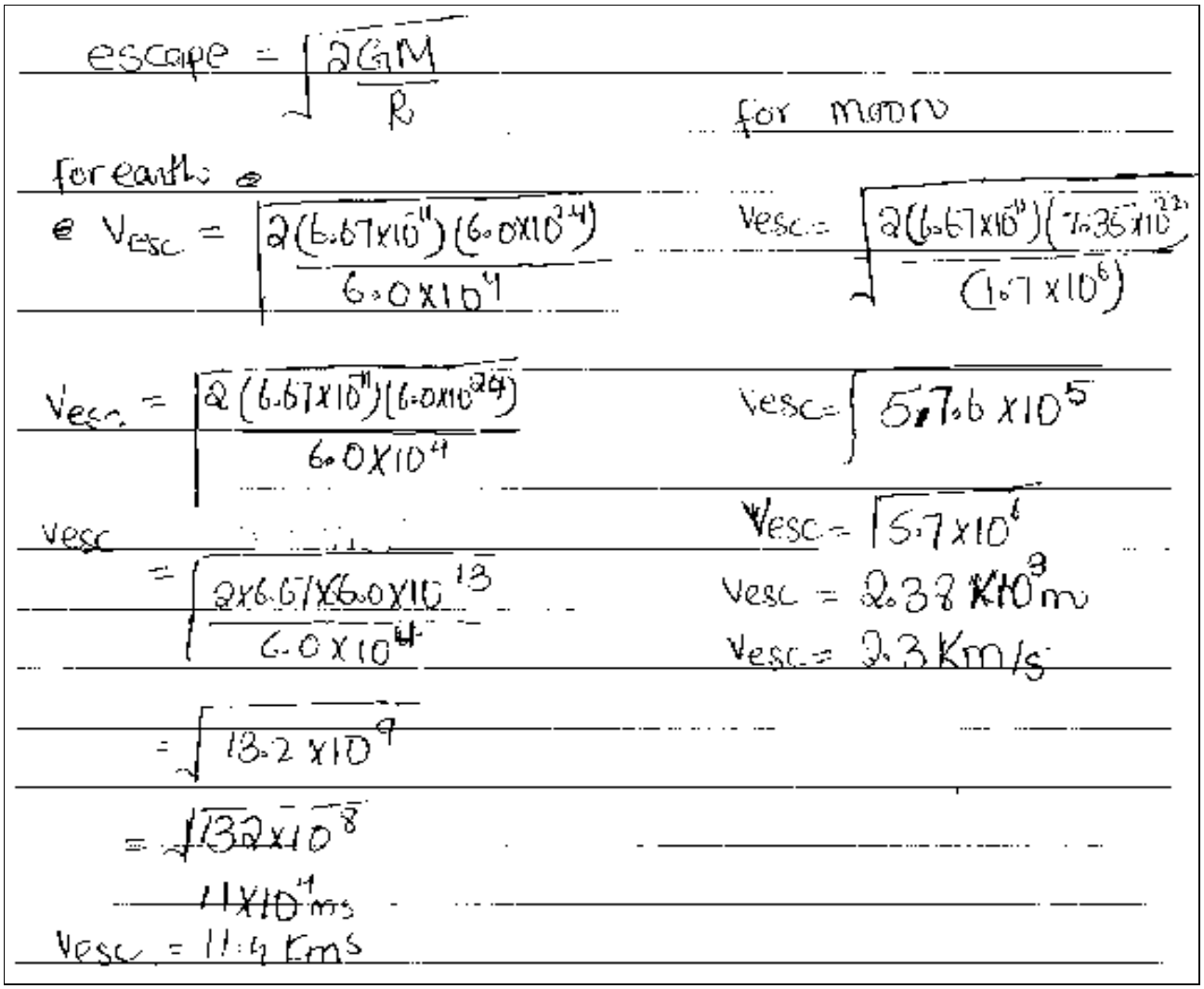
Question 12a:

Calculate the escape velocities for the Earth and the Moon if their masses are 6.0×10^{24} kg and 7.35×10^{22} kg with radii 6.4×10^{6} m and 1.7×10^{6} m respectively.

(**Note**: The value of Universal Gravitational Constant is 6.67×10^{-11} N. m^2/kg^2. Show your answer in kilometre per second.)

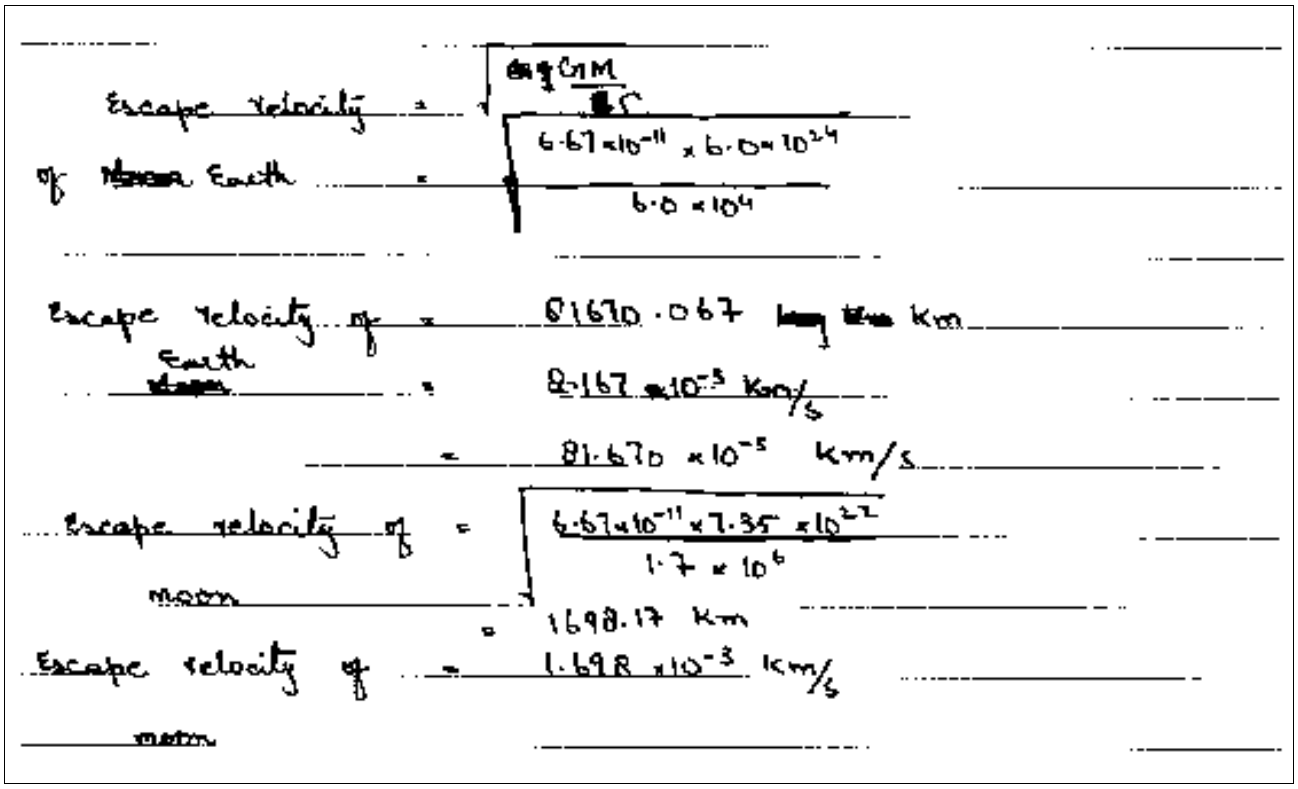
Better responses correctly calculated the escape velocities for the Earth and the Moon by extracting the data and using the correct formula of escape velocity, i.e. $V_{esp} = \sqrt{2GM_m/R_m}$. Some of the candidates used another formula for calculating the escape velocity for the Moon which was also correct. The formula was $V_{esp} = \sqrt{2gR}$ that led to the correct answer.

Example:



Weaker responses showed that candidates misunderstood the question. Few low-scoring candidates first used the incorrect formula for escape velocity and mixed it up with the escape velocities for the Earth and the Moon which showed their lack of understanding of the given concept.

Example:



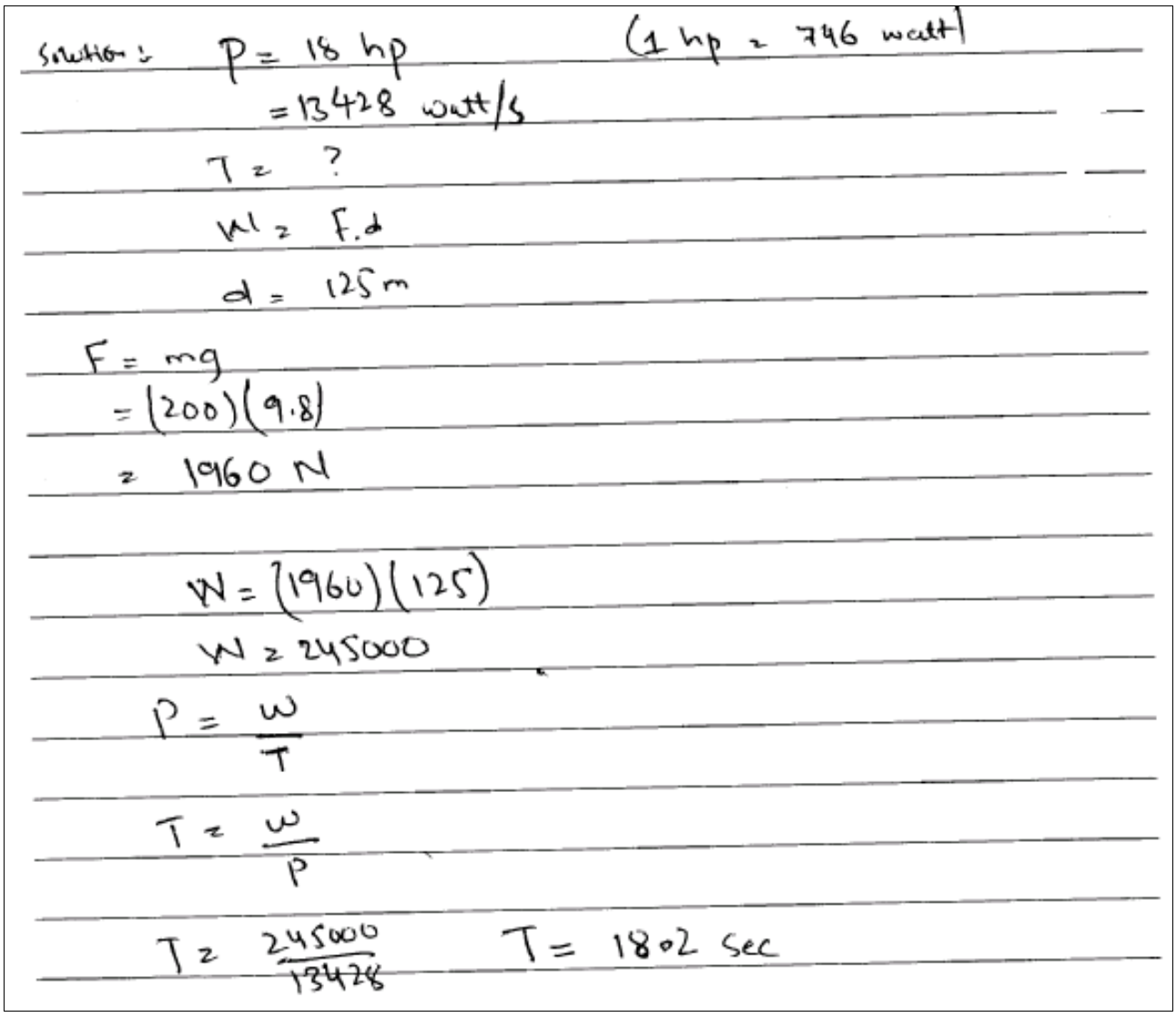
Question 12b:

An 18 horse power (hp) motor raises 200 kg of water through a height of 125 m. Calculate the time (in seconds) that would be needed to raise the water to this height.

(**Note**: The acceleration due to gravity 'g' is 9.8 m/s^2.)

Better responses correctly calculated the time that would be needed to raise the water to the required height. The high-scoring candidates extracted the correct data, converted the horse-power into watt, used the correct formula of power P = F × d/ t for calculating the time.

Example:



Weaker responses failed to calculate the time that would be needed to raise the water to the required height. In such responses, candidates extracted the correct data but were unable to apply the correct formula and used W = mgh for calculating the time.

Example:

b). mass = 200kg

height = 125m

$T = ?$

$g = 9.8 m/s^2$

$P.E = \rho g h \quad \therefore \rho = \frac{m}{V}$

$V = mgh$

$V = (200)(9.8)(125)$

$V = 1960(125)$

$\boxed{V = 245000\ ms^{-2}}$

Majority of the candidates attempted question **13a** and they scored well in this part.

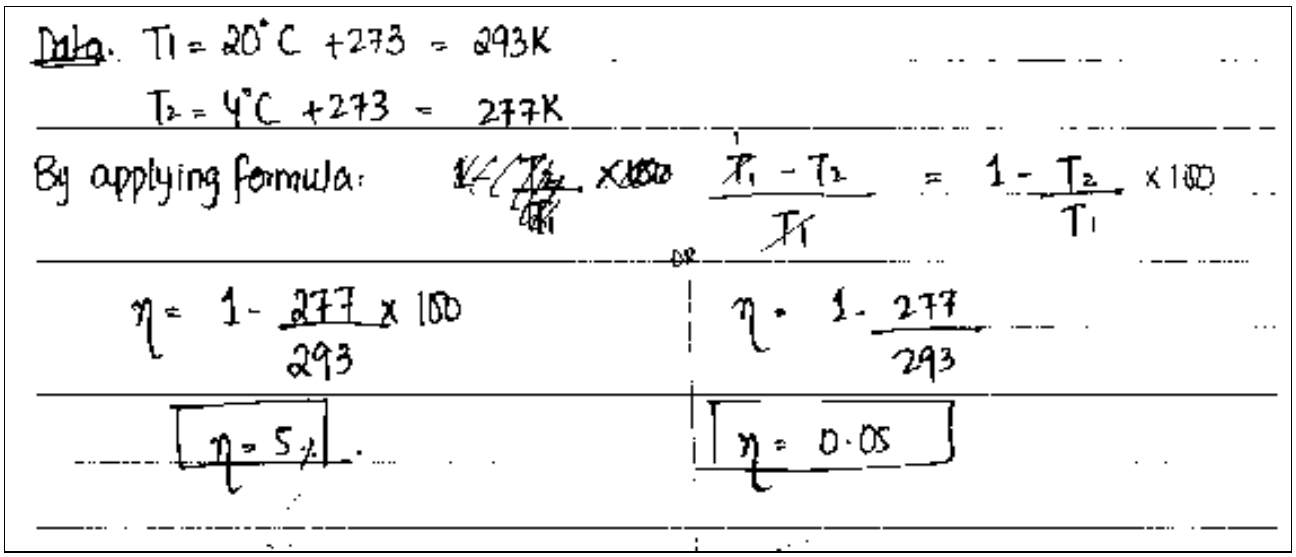
Question 13a:

Oceanologists are discussing about the difference in the temperature of water at the surface of the ocean and the bottom of the ocean for operating a heat engine.

If they observed 20°C as high temperature and 4°C as low temperature, then calculate the efficiency of the heat engine.

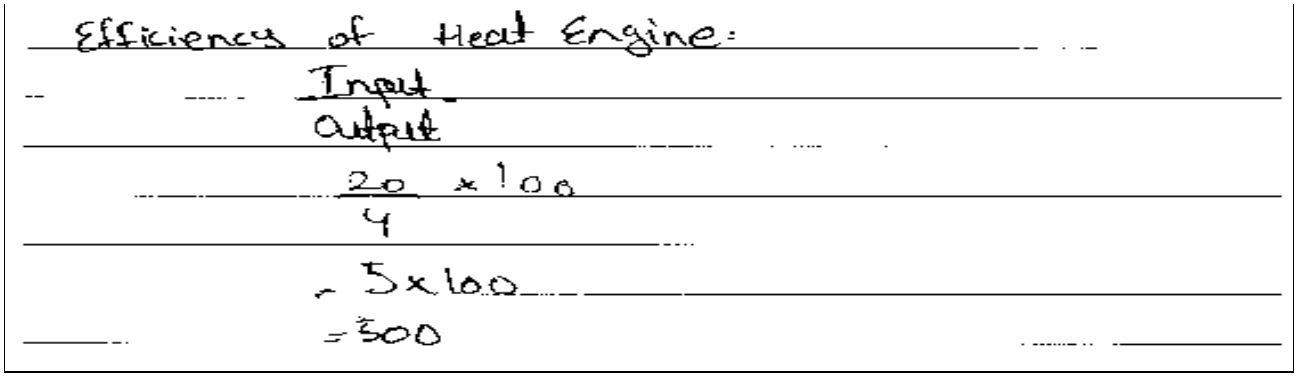
Better responses correctly calculated the efficiency of the heat engine. The high-scoring candidates correctly converted temperature from centigrade to kelvin, used the correct formula of efficiency, i.e. $\eta = T_1 - T_2 / T_1$ to calculate the answer.

Example:



Overall, very few candidates made mistakes in this question, almost all the candidates who attempted this question, attempted very well. However, a few w*eaker responses* showed that candidates were unable to apply the correct formula of efficiency and wrote input/ output that calculated the wrong efficiency of heat engine.

Example:



Question 13b:

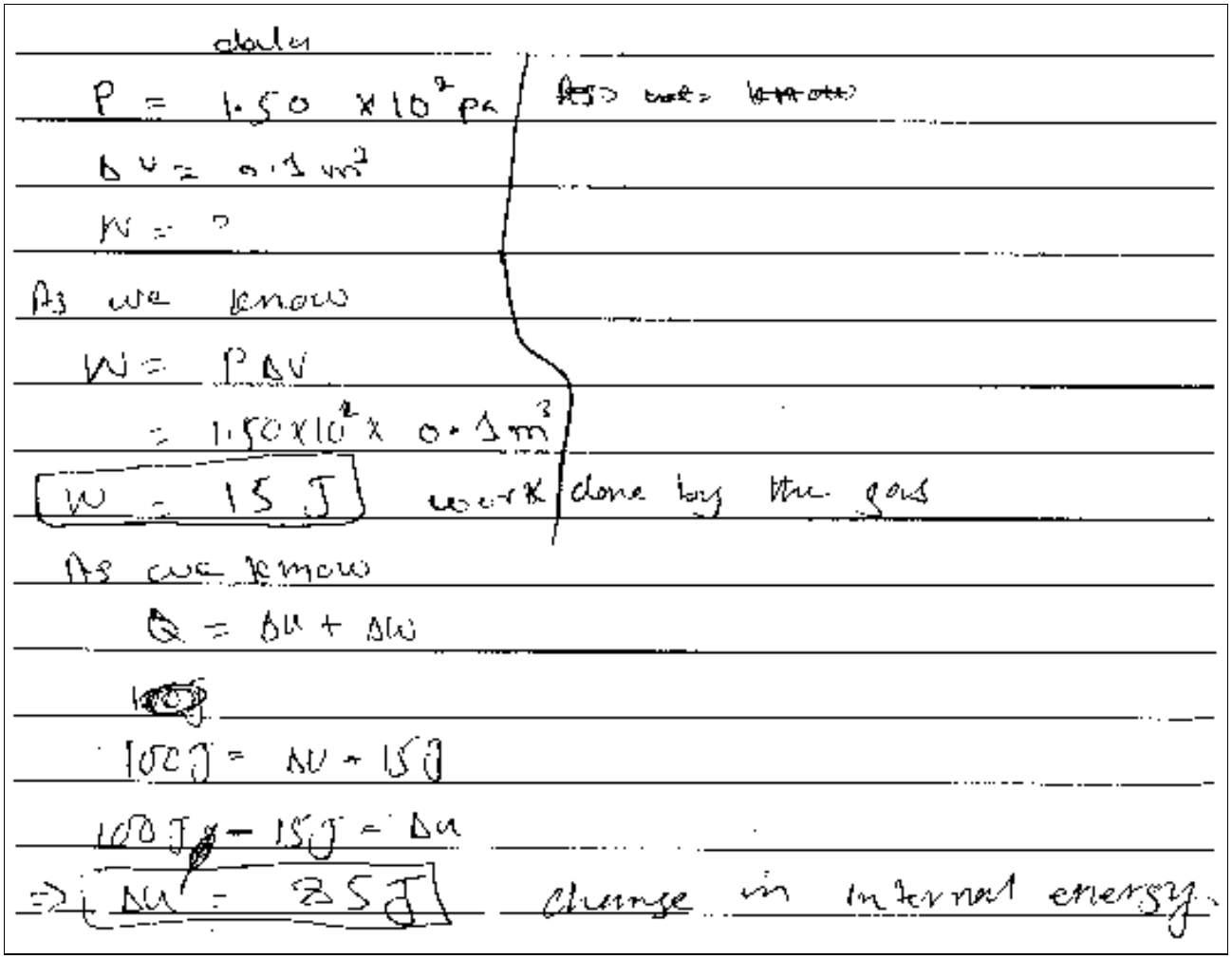
100 J of heat is added in the system of a gas at constant pressure of 1.50×10^2 Pa. If the system experiences a change in volume of 0.1 m^3, then calculate the

i. work done by the gas.

ii. internal energy change in the process.

Better responses correctly calculated the work done by a gas and internal energy in the given process. In part (i), candidates applied the correct formula, i.e. $W = P\Delta V$ and calculated the work done in thermodynamics system, W = 15 J. In part (ii), high-scoring candidates correctly calculated the internal energy by using the formula $\Delta U = Q - W$.

Example:



Weaker responses revealed that candidates were unable to provide all the relevant calculation. They neither wrote the correct formulae of work done in thermodynamics system and internal energy nor calculated the correct values. Some of the candidates wrote the formula of work done of kinematics, W = F.d and compared it with the heat, pressure and volume of a gas.

Example:

Data

(i) Heat added = 100 J

Preassure = 1.50×10^2 Pa

Volume in change = $0.1 m^3$ = 0.47m.

(i) Work done = F.d

= $1.50 \times 10^2 \times 0.47$

= 150×0.47

= 70.5 Nm.

(ii) Internal Energy change in process.

~~let suppose internal energy change for 2S.~~

So = $\frac{1.50 \times 10^2 \times 0.47}{2S}$ + 100 J

* = 70.5 J + 100 J

= 170.5 J